

- 6 (a) The electric force per unit positive charge experienced by a small stationary test charge placed at that point. **B1**

(b)(i) to the right / from the left / from A to B / in the same direction as electron velocity **B1**

(ii) $v^2 = u^2 + 2as$

$$a = (1.5 \times 10^7)^2 / (2 \times 2.0 \times 10^{-2}) \quad \text{C1}$$

Other alternative calculations for the C1 mark:

e.g. $a = 1.5 \times 10^7 / 2.67 \times 10^{-9}$

e.g. $a = [(1.5 \times 10^7 \times 2.67 \times 10^{-9}) - 2.0 \times 10^{-2}] \times [2 / (2.67 \times 10^{-9})^2]$

e.g. $a = (2.0 \times 10^{-2} \times 2) / (2.67 \times 10^{-9})^2$

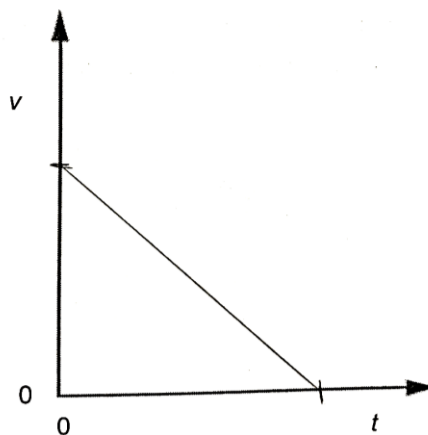
$$= 5.6 \times 10^{15} \text{ m s}^{-2} \quad \text{A1}$$

(iii) $E = F / Q$

$$= (9.11 \times 10^{-31} \times 5.6 \times 10^{15}) / 1.6 \times 10^{-19} \quad \text{C1}$$

$$= 3.2 \times 10^4 \text{ V m}^{-1} \quad \text{A1}$$

(c) straight line with negative gradient starting at an intercept on the v -axis and ending at an intercept on the t -axis. **B1**



7 (a) Angular velocity is the rate of change of angular displacement of a radius joining